

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We construct a graph-completed four-port symplectic self-return map and an autonomous Hamiltonian suspension whose work layer covers every regular incoming port point. The integrated mechanical work is therefore exactly the branch-current cocycle on the full Liouville path law. A finite trace-jet Banach scale gives an all-order physical moving-seam response calculus and its common-root finite-difference realization. Identification of a valuation coefficient with the mechanical conjugate field is stated precisely as a canonical-cocycle compatibility theorem.

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1. ROUND-SIX POSITIVE CLOSURE: FULL-PORT SUSPENSION, EXACT WORK, AND PHYSICAL RESPONSE

The construction below uses one graph-completed self-return section from the outset. In particular, no outgoing rectangle is identified with a boundary of the wrong type, and the work layer is traversed while the four incoming ports are genuinely disjoint. Thus the mechanical work equals the branch

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current on the whole regular Liouville section, not only on a smaller port core.

1.1. The graph-completed port section. Let

$$0 = s_0(a) < s_1(a) < \cdots < s_4(a) = 1, \quad I_i(a) = (s_{i-1}(a), s_i(a)),$$

with widths $p_i(a) > 0$. Put

$$\widehat{\Sigma}_a = \bigsqcup_{i=1}^4 \overline{I_i(a)}^g \times [0, 1],$$

where every endpoint of an interval has its two oriented graph-completion copies. A point of $\widehat{\Sigma}_a$ therefore carries its incoming port label even at a seam. On the interior of port i , define

$$Q_i(q, p) = \frac{q - s_{i-1}(a)}{p_i(a)}, \quad P_i(q, p) = s_{i-1}(a) + p_i(a)p.$$

Let $j = j(Q_i)$ be the unique next-port label for which $Q_i \in I_j(a)$, with the corresponding one-sided label on the graph completion, and set

$$F_a(i, q, p) = (j, Q_i(q, p), P_i(q, p)).$$

Lemma 1.1 (One-section exact symplectic recutting). *The map $F_a : \widehat{\Sigma}_a \rightarrow \widehat{\Sigma}_a$ is a bijective, stratum-preserving piecewise C^∞ diffeomorphism. On each stratum it preserves $dq \wedge dp$ and has exact primitive*

$$S_{a,i}(q, P) = \frac{(q - s_{i-1}(a))(P - s_{i-1}(a))}{p_i(a)}.$$

In affine fixed-port coordinates it depends C^m on a whenever the widths do. Its symbolic factor is the full four-symbol shift.

Proof. On each rectangle $I_i(a) \cap Q_i^{-1}(I_j(a)) \times (0, 1)$, the formula is affine and its Jacobian matrix is

$$\begin{pmatrix} p_i(a)^{-1} & 0 \\ 0 & p_i(a) \end{pmatrix},$$

so the determinant is one and the symplectic form is preserved. The inverse is obtained by reading the previous horizontal strip label from the second coordinate and applying the inverse affine formula. Oriented endpoint copies make both label assignments single valued at every seam, hence the branch maps paste to a bijection of the graph completion. The displayed generating function gives $Q = \partial_P S_{a,i}$ and $p = \partial_q S_{a,i}$. After the affine changes $u = (q - s_{i-1})/p_i$, $v = p$, every branch has fixed domain and its parameter dependence is carried by smooth coefficients. Since Q_i ranges over $(0, 1)$, every next label occurs on a nonempty open rectangle. $\square$

1.2. Autonomous Hamiltonian suspension. Define the mapping torus

$$\mathcal{M}_a = (\widehat{\Sigma}_a \times [0, 1]_\tau \times \mathbb{R}_E) / ((x, 1, E) \sim (F_a x, 0, E)).$$

On every corner stratum set

$$\Omega_a = dq \wedge dp + dE \wedge d\tau, \quad H_a = E.$$

Theorem 1.2 (Full graph-completed Hamiltonian return flow). *The quotient $\mathcal{M}_a$ is a Hausdorff symplectic manifold with corners. Its Hamiltonian flow is complete, has return time one on $\widehat{\Sigma}_a \times \{0\} \times \{E_0\}$, and has first-return map exactly F_a . Moreover it admits a smooth autonomous impact-network realization obtained by replacing each affine branch by a compactly supported Hamiltonian suspension in a disjoint port cobordism; the replacement is symplectically conjugate to the mapping-torus flow near the return section and preserves section flux and return time.*

Proof. The graph of the stratum-preserving homeomorphism F_a is closed in the compact graph-completed section. Therefore the equivalence relation defining the mapping torus is closed and the quotient is Hausdorff. Corner charts are mapping cylinders of smooth diffeomorphisms between finite orthants. Since $F_a^*(dq \wedge dp) = dq \wedge dp$, the two boundary traces of Ω_a agree and the form descends. It is nondegenerate on every stratum. The Hamiltonian vector field of E is ∂_τ , so one unit of time reaches the glued boundary and applies F_a .

For the smooth network realization, use the exact branch primitive from Lemma 1.1. On a disjoint Darboux copy of each incoming port, the translation, diagonal symplectic dilation, and target translation are time-one maps of explicit linear Hamiltonians. Concatenate them in three disjoint time windows. Because the port components are disjoint, the four Hamiltonians have disjoint spatial support. A relative extension across the collar is obtained by integrating the compactly supported exact difference of the branch primitives; the resulting Hamiltonian vector field vanishes on both collar boundaries. The recutting hub is the coordinate identity on the graph completion and is inserted only after the branch cobordisms. Standard autonomization with the pair (τ, E) gives an autonomous flow. The conjugacy is the identity on the return section, hence preserves the return map, time, and Liouville flux. $\square$

1.3. Exact mechanical work on every port point. Split one flight into a work layer $0 < \tau < 1/3$, a branch layer $1/3 < \tau < 2/3$, and a recutting layer $2/3 < \tau < 1$. In the work layer the phase space is the disjoint union of the four incoming port cylinders. Choose $\rho \in C_c^\infty((0, 1/3))$ with $\int_0^1 \rho(\tau) d\tau = 1$. Define

$$\alpha_a|_{\{i\} \times \overline{I}_i^g \times [0, 1]} = \epsilon_i \rho(\tau) d\tau.$$

Extend it by zero through the branch and recutting layers.

Proposition 1.3 (Full-port work form). *The one-form α_a is globally smooth on the port cobordism and descends to the autonomous suspension. For every regular section point in port i , including points arbitrarily close to a seam,*

$$\int_0^1 \alpha_a(X_{H_a}) dt = \epsilon_i.$$

Consequently the mechanical work of the first n returns is exactly

$$W_n(x) = \sum_{k=0}^{n-1} \epsilon_{\iota(F_a^k x)}$$

on the full regular Liouville path law. The equality is not restricted to a port core and remains true when consecutive labels differ.

Proof. Before the branch and recutting layers, the four port components are disjoint closed manifolds with corners. A constant coefficient on each component is therefore smooth; no cutoff in the section coordinate is required. The form vanishes near the temporal interfaces, so its traces agree under every later gluing. Along the work layer, $X_{H_a} = \partial_\tau$ and the port label is constant. Integration gives $\epsilon_i \int \rho = \epsilon_i$. Concatenating return segments proves the additive formula. $\square$

1.4. Physical trace spaces and all-order shape response. Let $\mathcal{C}_a$ be the finite cell decomposition of $\hat{\Sigma}_a$ by the current and next-port labels. For integers $r \geq 0$, $m \geq 0$, and $0 < \eta < 1$, define

$$\mathcal{B}_a^{m,r,\eta} = \bigoplus_{C \in \mathcal{C}_a} C^{m+r,\eta}(\overline{C}) \oplus \bigoplus_{\gamma \in \partial \mathcal{C}_a} \bigoplus_{\ell=0}^r C^{m+r-\ell,\eta}(\gamma),$$

with the compatibility relation that the boundary coordinates are the oriented one-sided trace jets of the cell functions. Affine fixed-cell maps identify these spaces with one Banach space $\mathcal{B}_*^{m,r,\eta}$. The physical transfer operator $\mathcal{L}_a$ acts by inverse branches on cell coordinates and by the induced triangular trace-jet action on boundary coordinates.

Lemma 1.4 (Bounded seam flux and shape derivative). *For $r \geq m$, the conjugated operators $J_a \mathcal{L}_a J_a^{-1}$ are C^m in $\mathcal{L}(\mathcal{B}_*^{m,r,\eta}, \mathcal{B}_*^{0,r-m,\eta})$. Every seam flux and every derivative of a seam flux up to order m is a bounded functional on the corresponding trace coordinate. The first physical derivative is*

$$D_a \langle G, \nu_a \rangle = \langle D_a^{\text{bulk}} G, \nu_a \rangle + \langle (D_a \mathcal{L}_a) S_a Q_a G, \nu_a \rangle + \sum_{\gamma} \dot{s}_\gamma(a) \langle [G]_\gamma, j_\gamma \rangle.$$

Proof. In fixed-cell coordinates the branch maps and Jacobians are smooth rational functions of the positive widths $p_i(a)$. A parameter derivative either differentiates a bulk coefficient or moves one face. The distributional shape derivative of a moved face is its oriented trace; subsequent derivatives produce the higher trace jets already included in the definition of $\mathcal{B}_a^{m,r,\eta}$. Because the cell complex is finite, the sum of all bulk and trace blocks is bounded uniformly on compact parameter sets. The spectral gap on the

finite-branch expanding factor gives the reduced resolvent $S_a = (I - \mathcal{L}_a)^{-1} Q_a$. Differentiating the invariant pairing and applying the Reynolds transport formula yields the displayed identity. $\square$

Theorem 1.5 (All-order physical and symbolic response). *For every m , observables in $\mathcal{B}_a^{m,m,\eta}$ have parameter derivatives through order m . Each derivative is an absolutely convergent finite sum of words in the bulk operator derivatives, reduced resolvents, projectors, and trace-jet fluxes. The same quantities are obtained from centered finite differences under the common uniform-root coupling. The symbolic Bernoulli response is the factor of this physical formula obtained by discarding zero jump traces.*

Proof. Iterate Lemma 1.4. At derivative order k , at most k trace derivatives are created, all of which are coordinates of $\mathcal{B}_a^{m,m,\eta}$. Between insertions, the centered transfer operator contracts by a uniform factor, so every time-index sum is geometric. Taylor's formula in the fixed-cell Banach space gives convergence of centered finite differences. Under the common-root coupling, crossing a moving seam produces precisely the oriented trace term, while points not crossing a seam contribute the bulk derivative. Dominated convergence identifies the two forms of the response. $\square$

1.5. Preparation and conditional calibration. The multinomial current preparation on the common symbolic factor has canonical likelihood

$$\frac{d\mathbb{P}_\theta}{d\mathbb{P}_0} = \exp\{\theta W_n - n[Q(\theta) - Q(0)]\}.$$

Theorem 1.6 (Conditional mechanical–valuation identification). *A smooth cash-additive, law-invariant, strongly time-consistent certainty equivalent generated by one utility is affine or entropic with a constant coefficient ϑ . It has $\vartheta = \theta$ exactly when its likelihood cocycle is required to equal the canonical mechanical work cocycle in the same units. Mechanics alone does not select the coefficient of an unrelated preference functional.*

Proof. Cash additivity gives the translation functional equation for the utility; two differentiations make absolute risk aversion constant. Strong time consistency makes that constant independent of the time split. Equality of the two normalized likelihoods on the nonconstant random variable W_n forces $(\vartheta - \theta)W_n$ to be constant, hence $\vartheta = \theta$. The converse is immediate. $\square$

Theorem 1.7 (Round-six A1 closure). *The four-port family possesses one well-defined graph-completed symplectic return map, an autonomous Hamiltonian suspension, a smooth full-port work form whose integral is exactly the branch current on the complete regular Liouville section, and a fully typed all-order physical response calculus. The valuation identification is precisely the conditional compatibility statement of Theorem 1.6.*

Proof. Combine Lemma 1.1, Theorem 1.2, Proposition 1.3, and Theorems 1.5 and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

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